

Technology Stack

Date	27 june 2026
Team ID	SWTID-2026-6540
Project Name	OptiCrop: Smart Agricultural Production Optimization Engine
Maximum Marks	2 Marks

Technology Stack Details

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	Frontend / Client-Side	<i>HTML, CSS</i>	HTML is used to create the structure of the OptiCrop web application. It defines elements such as forms, input fields, buttons, tables, and pages where users enter soil and environmental data and view crop recommendations. CSS is used to design and style the HTML pages, making the application visually attractive and easy to use.

2	Backend / Server-Side	<i>[Specify technology]</i> PYTHON,FLASK	Python is the primary programming language used to develop the OptiCrop application and implement the machine learning model. It is widely used in AI and data science because of its simple syntax and powerful libraries. Flask is a lightweight Python web framework that connects the machine learning model with the web interface. It receives user input, processes it using the ML model, and returns the predicted crop.
3	Database / Data Storage <i>(e.g., MySQL, MongoDB, PostgreSQL)</i>	None (CSV Dataset)	OptiCrop uses a CSV dataset to store agricultural data. The dataset contains soil nutrients, environmental parameters, and crop information, which are used to train the machine learning model. During prediction, the trained model (model.pkl) processes user input directly without requiring a database.

4	Cloud / Hosting / Deployment	LOCAL HOST	
5	Version Control & CI/CD	<i>GitHub</i>	Stores the project code securely in an online repository. Tracks changes and maintains version history. Enables collaboration among team members.
	<i>(e.g., GitHub, GitLab, Jenkins)</i>		
6	Third-Party APIs / Other Tools	No external APIs are used.	